

GlucoTwin — Architecture

A Type 2 Diabetes Digital Twin · Data flow & component diagram

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MERI College of Engineering and Technology
Happiest Health Digital Twin Challenge 2026

LAYER 1 — SYNTHETIC DATA GENERATION



Static EHR Generator

200 synthetic patients · age, sex, BMI, HbA1c, medication, hypertension, genetics, fasting glucose, eGFR · correlated fields (BMI ↔ HbA1c, ADAG equation) · fixed seed

data/ehr.csv



Dynamic Wearables Generator

14 days × 5-min intervals · glucose, HR, HRV, steps, sleep stages, meal carbs, activity · physiological glucose model (meal absorption, dawn, exercise, sleep carryover)

data/wearables.csv · 806K samples



LAYER 2 — FEATURE ENGINEERING (STATIC + DYNAMIC FUSION)

33-Feature Fusion · 6-hour lookback window

Current glucose · 15/30/60-min rate of change · rolling mean/std/min/max · time since last meal · last meal carbs · rolling steps/HR/HRV · sleep hours/deep% · hour of day · + age, sex, BMI, HbA1c, medication, genetics, eGFR, insulin sensitivity

33
features

Label: max glucose in next 120 min > 180 mg/dL

Split: patient-wise 70/15/15 (no leakage)

10,200 labeled windows



LAYER 3 — MODEL TRAINING

Logistic Regression

Baseline · L2 regularized · gradient descent · ROC-AUC 0.948

GBDT Classifier ★

80 trees · depth 4 · histogram splits · Newton leaves · log loss · ROC-AUC 0.981 · F1 0.940

GBDT Regressor

Glucose at +30/60/120 min · MAE 4.5 / 5.7 / 7.0 mg/dL · 95% confidence band



LAYER 4 — MODEL ARTIFACTS + INFERENCE

Serialized JSON models + Tree-Interpreter (SHAP-style)

gbdt-classification.json · gbd-t-regression.json · logreg.json · metrics.json · Per-prediction feature contributions in log-odds space → human-readable "why" reasons

ml/*.json



LAYER 5 — API + TWIN SIMULATOR

REST API (Next.js route handlers)

GET /api/patients	GET /api/patients/:id
GET /api/patients/:id/timeseries	GET /api/patients/:id/risk
POST /api/patients/:id/whatif	GET /api/model/metrics
GET /api/stream/:id (SSE)	GET /api/health

 Twin Simulator (What-if)

Physiological model: meal absorption + insulin clearance + exercise effect + medication adjustment → baseline vs simulated 3-h glucose curve + change in spike risk



LAYER 6 — DOCTOR DASHBOARD

Single-page doctor dashboard (React + Recharts + Tailwind)

Patient List · Digital Twin View (glucose + prediction + risk gauge + SHAP + EHR + wearables + alerts) · What-if Simulator · Model Insights · About

